25901629 **A**

Reg.No : .....

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Multi-Disciplinary Courses (MDC) - MG2MDCCHE100 - DAIRY CHEMISTRY

(2024 ADMISSION ONWARDS)

Duration: 45 Minutes **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple choice questions

Answer all questions

Each question carries **1** marks

1. What type of fat is mainly found in milk? [U] / [CO1]

   a). Trans fat b). Saturated fat

   c). Unsaturated fat d). Polyunsaturated fat

2. Which process extends milk's shelf life? [U] / [CO1]

   a). Boiling b). Pasteurization

   c). Freezing d). Fermentation

3. Colostrum is rich in which type of vitamin? [U] / [CO1]

a). Vitamin A b). Vitamin C c). Vitamin B12 d). Vitamin D

4. Name the administrative head of FSSAI [K] / [CO1]

a). Ministry of Livestock and Poultry

b). Ministry of Health and Family welfare

c). Ministry of Food and Cosmetics

d). Ministry of Child Health Care

5. The acidity of milk is expressed in terms of: [U] / [CO1]

a). pH b). Titratable acidity

c). Both pH and Titrable acidity d). None of the options are correct

6. AGMARK is primarily concerned with: [K] / [CO1]

a). Ensuring the quality of food products

b). Ensuring food safety standards

c). Regulating food prices

d). Encouraging exports of food

7. Choose the substance which cannot be used as a preservative for milk [K] / [CO1]

a). Boric acid b). Benzoic acid c). Formaldehyde d). Sodium hydroxide

8. Give the colour when milk adulterated with starch is treated with iodine [K] / [CO1]

a). Pink b). Yellow c). Blue d). Red

9. Which milk processing method is used to destroy pathogens and extend milk's shelf life? [U] / [CO2]

a).Homogenization b). Filtration c).Pasteurization d). Clarification

10. The Process of Removing unwanted particles and cream from milk is [K] / [CO2]

a). Dehydration b). Clarification c).Decolourisation d). neutralization

11. Find the factor which does not increases the yellow colour of the milk [U] / [CO1]

a). Large fat globules b). Decreased green feed

c). High fat percentage d). Increased green feed

12. Explain how increase in homogenisation affects the viscosity of the milk [U] / [CO1]

a). Increases b). Decreases

c). Has no effect d). Cannot predict

13. Which immune factor is found in high amounts in colostrum? [K] / [CO1]

a).Immunoglobulins b). Carotenoids c). Amino acids d). Enzymes

14. What is the main purpose of lactose in milk? [K] / [CO1]

a). Provides energy b). Enhances flavor

c). Stabilizes fat d). Preserves milk

15. According to the PFA Act, 1954, the term "adulterated food" refers to food: [U] / [CO1]

a). That is unsafe for consumption

b). That has been tampered with to increase quantity or weight

c). That is prepared in an unclean environment

d). All options are correct

16. Which of the following is true about LTLT pasteurization? [K] / [CO2]

a). It is typically done at 63°C for 15-20 minutes

b). It is used at a temperature of 85°C for 5 minutes

c). It is not effective in killing all bacteria

d). It is more suitable for high-volume production

17. The component of milk which serves as a solvent for other nutrients. [K] / [CO1]

a). Fat b). Protein c).Carbohydrate d). water

18. Which macronutrient is most abundant in condensed milk? [K] / [CO3]

a). Fat b).Carbohydrates c). Protein d). Fiber

19. What is butter? [K] / [CO4]

a). A type of cheese made from milk b). A dairy product made by churning cream

c). A solidified form of milk d). A product made from vegetable oil

20. What is the main characteristic of vegetable-toned milk? [U] / [CO3]

a). It is made from cow's milk and vegetables b). It is a milk product with a mixture of vegetable oils

c). It is exclusively made from plant-based ingredients d). It is a non-dairy milk with no vegetable oils

21. What is added to reduce fat content during standardization? [K] / [CO3]

a). Skim milk b). Cream

c). Water d). Sugar

22. What is a drawback of flavored milk? [K] / [CO3]

a). Higher protein content b). High added sugar content

c). Added artificial sweeteners d). Increased saturated fat

23. Why is imitation milk consumed? [U] / [CO3]

a). Lower cost b). Lactose intolerance or veganism

c). Higher fat content d). Added probiotics

24. Choose the protein found in cream. [U] / [CO4]

a). Globulin b). Casein c). Lactalbumin d). Ovalbumin

25. Milk which is obtained by adding one or more vitamins is called [U] / [CO3]

a). Vitaminised milk b). Vegetable toned milk

c). Flavoured milk d). Standardised Milk

26. Which microorganisms are mainly responsible for milk fermentation? [U] / [CO4]

a). Fungi b). Viruses c). Lactic acid bacteria d). Yeast

27. Milk obtained by the addition of water and skim milk powder to the whole milk is called [U] / [CO3]

a). Toned milk b). Imitation milk

c). Standardised milk d). Homogenised milk

28. In the production of desi butter, which of the following is typically used? [U] / [CO4]

a). Whole milk b). Milk powder c). Cream d). Skimmed milk

29. Explain how Khoa is made. [U] / [CO4]

a). Boiling milk to remove water b). Fermenting milk with bacteria c). Adding vinegar to milk d). Mixing milk with sugar

30. The less common drying method which gives a coarser product. [U] / [CO4]

a). Spray drying b). Drum drying

c). Cooling d). All options are correct

31. Which process is often paired with homogenization but serves a different purpose? [K] / [CO3]

a). Fermentation b). Pasteurization

c). Freezing d). Sweetening

32. The most celebrated Indian sweet originating from West Bengal is [U] / [CO4]

a). Cream b). Kulfi c). Rasgulla d). Gulab Jamun

33. Explain the higher fermentation temperature typically used for yogurt production? [U] / [CO4]

a). 20-25°C b). 30-35°C c). 42-45°C d). 50-55°C

34. Which is a common practice for detecting the presence of animal fats in ghee? [K] / [CO4]

a). Gas chromatography b). Iodine solution test

c). Sensory analysis (smell and taste) d). Acidity test

35. Which of the following is an effective preventive measure against rancidity in food products? [U] / [CO4]

a). Reducing the temperature at which the food is stored

b). Increasing the humidity around food

c). Storing food in metal containers

d). Keeping food exposed to sunlight

[1x35 = 35]

25901629 **B**

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